

Inglés I

Listening:

Margaret Hamilton and the Apollo XI mission

Margaret Hamilton: The First Software Engineer.

Watch a [video](#) about Margaret Hamilton, a programming pioneer whose work was fundamental to the Apollo mission. She was responsible for the creation of the term 'software engineer' and a pillar in the mission's ultimate success.

Pre-watching activity

Here are some terms that will appear in the video. Match them to their corresponding meanings:

1) Asynchronous	a) to give someone a specific job, task, or responsibility
2) Assign	b) Refers to processes or data transmissions that run independently without waiting for a shared clock signal or for another task to finish first.
3) Ensure	c) system, device, or plan designed to stop working safely or prevent disaster if a part breaks or malfunctions
4) Regardless	d) To send too much electrical current through a circuit
5) Warn	e) To make certain that something happens or is true

6) Fail safe	f) doing something in spite of difficulties, or without being affected or stopped by a specific factor or problem
7) To overload	g) tell someone about a possible danger, problem, or bad consequence in advance so they can avoid it

Now, let's watch a segment from the video. Listen to it and complete it with the missing words:

Watch the video from [this moment](#) on and complete the text with the missing words:

NASA looked to Draper and his group of 400 engineers to invent the first compact digital flight computer: the Apollo guidance computer. Using input from astronauts this device would be responsible for guiding, navigating and controlling the spacecraft. At a time when unreliable computers filled entire rooms the AGC needed to operate without any errors and fit in one cubic foot of space. Draper divided the lab into two (1)_____: one for (2)_____ hardware and one for (3)_____ software. Hamilton led the team that built the onboard flight software for both the command and lunar modules. This work , for which she coined the term 'software engineering' was incredibly high stakes: (4)_____ lives were on the line so every program had to be (5)_____. Margaret's software needed to quickly detect unexpected errors and recover from them in (6)_____. But this kind of adaptable program was (7)_____ to build since early software could only process jobs in a predetermined order. To solve this (8)_____, Margaret designed her program to be asynchronous, meaning the software's more important jobs would interrupt less important ones her team assigned every task a unique

priority to ensure that each job occurred in the correct order and at the right time, regardless of any (9) _____. After this breakthrough, Margaret realized her software could help the astronauts work in an asynchronous environment as well. She designed priority displays that would interrupt astronauts' regularly scheduled tasks to warn them of (10) _____. The astronaut could then (11) _____ with Mission Control to determine the best path forward. This marked the first (12) _____ flight software communicated directly and asynchronously with a (13) _____.

Now, let's watch the full video and then let's do the multiple-choice quiz below:

1) Margaret Hamilton was born in _____.

- a) Akron, Ohio
- b) Detroit, Michigan
- c) Erie, Pennsylvania
- d) Paoli, Indiana

2) 400 scientists including Margaret Hamilton worked on the _____ at MIT to help send people to the Moon.

- a) Apollo Guidance Computer
- b) Asynchronous Radio Detector
- c) Air Defense System
- d) Lunar Radar Computer

3) Margaret's computer software had to be _____, meaning the software's more important jobs could interrupt less important ones.

- a) Chronological
- b) Asynchronous
- c) Compact
- d) Synchronous

4) The _____ would interrupt astronaut's regularly scheduled tasks to warn them of emergencies. The astronaut could then communicate with Mission Control to determine the best path forward.

- a) Altimeter
- b) Chronometer
- c) Flight Director
- d) Priority Displays

5) Margaret Hamilton and other scientists across the country were working to achieve _____'s goal of landing a man on the Moon by the end of the decade.

- a) Charles Draper
- b) President John F. Kennedy
- c) Neil Armstrong
- d) Richard Nixon

Answer Key:

Pre-watching activity:

1: b 2: a 3: e 4: f 5: g 6: c 7: d

Fill-in-the-blanks video activity:

- 1) teams**
- 2) designing**
- 3) developing**
- 4) human**
- 5) perfect**
- 6) real time**
- 7) difficult**
- 8) problem**
- 9) surprises**
- 10) emergencies**
- 11) communicate**
- 12) time**
- 13) pilot**

Multiple-choice quiz:

- 1) d)**
- 2) a)**
- 3) b)**
- 4) d)**
- 5) b)**

Source: Ted-Ed lesson on Youtube

 NASA's first software engineer: Margaret Hamilton – Matt Porter & Margar...